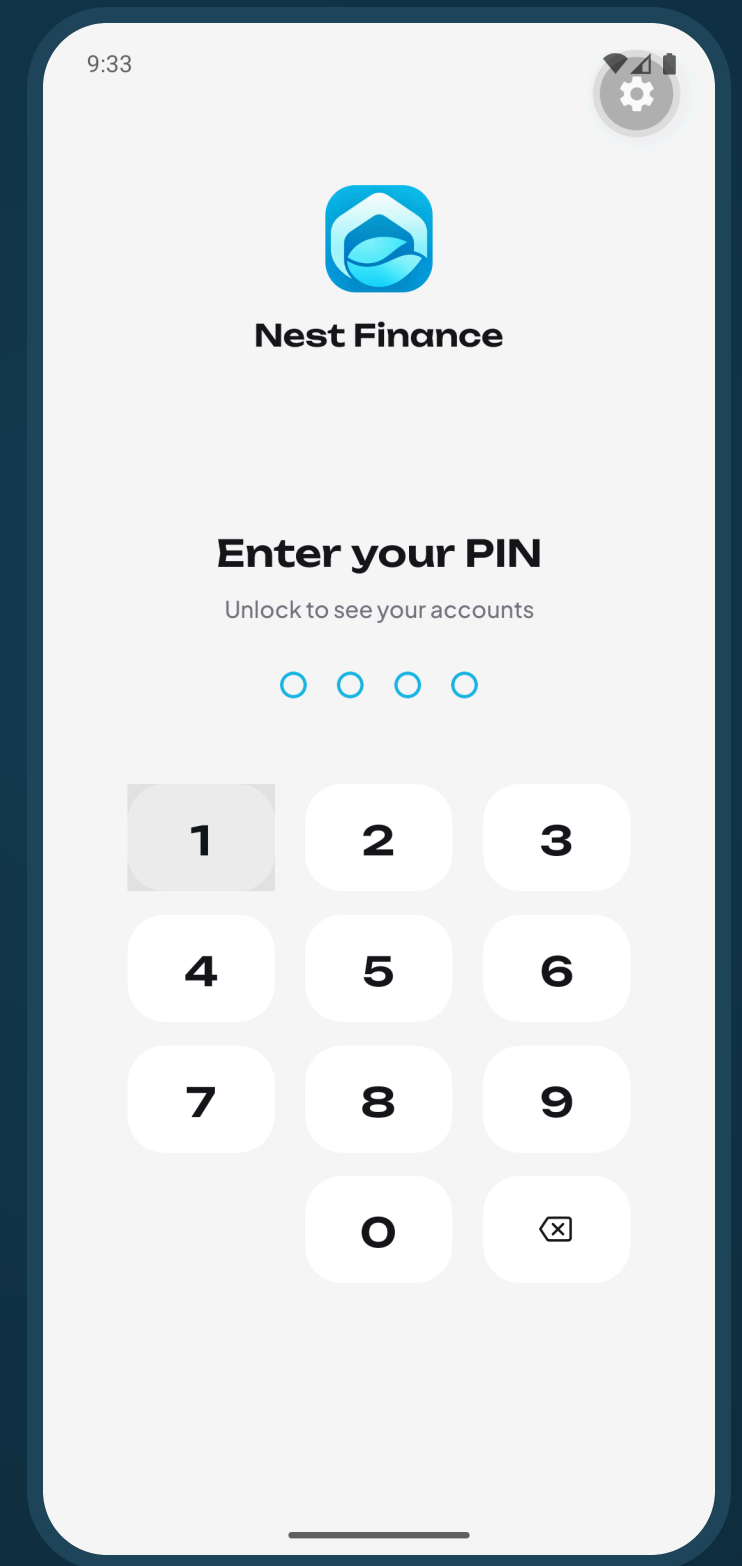




Nest Finance

Savings that can't be rushed.

A time-locked savings vault with a backup PIN and guardians, built for Solana Seeker.



THE PROBLEM

A Seeker in your hand says you hold crypto.

Forced to unlock

Someone next to you makes you open your wallet. Seed phrases and hardware keys don't help when they have you and your phone.

One bad signature

A drainer needs a single approval to empty a wallet.

No time to react

Crypto moves instantly. By the time anyone knows, the money is gone.

THE IDEA

Even with your real key, nobody can rush your savings.

Time-locked vault

Every withdrawal waits a delay you choose, enforced on-chain.

Backup PIN

Opens an ordinary wallet while your savings freeze in the background.

Guardians

People you trust can stop a withdrawal inside the window.

HOW IT WORKS

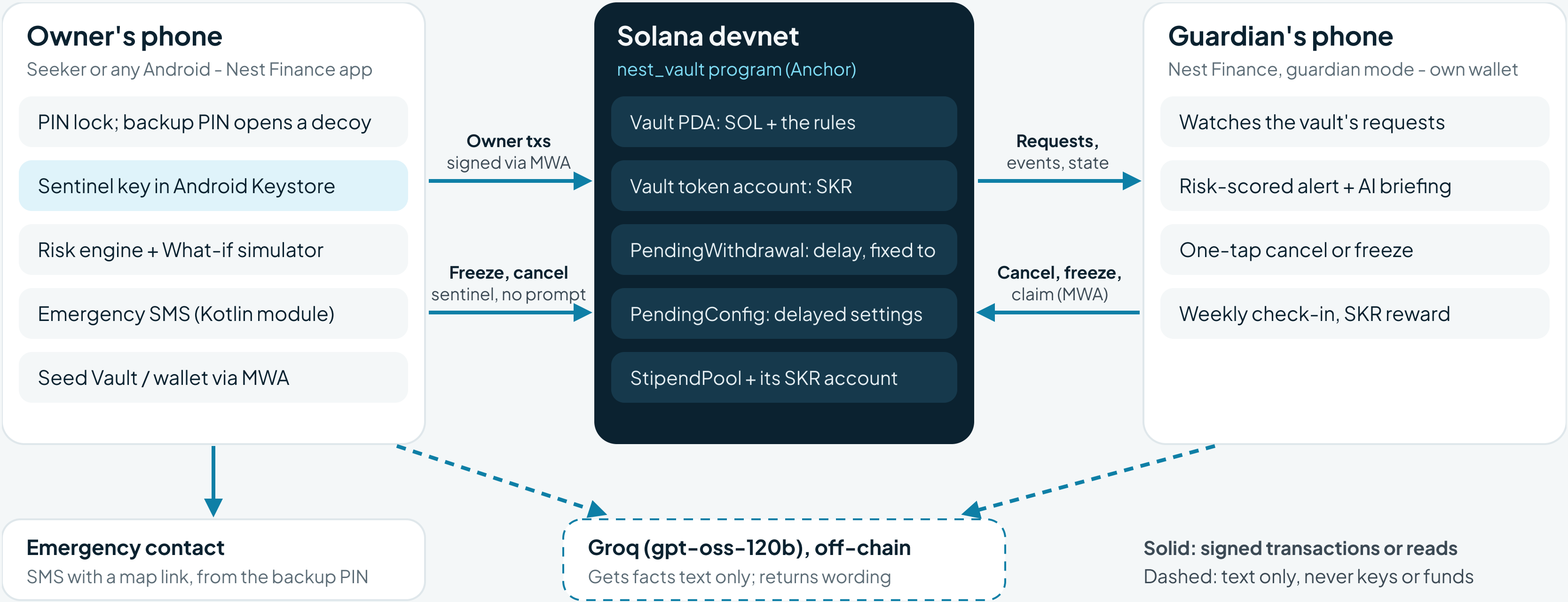
The program enforces the rules, not the app.

- Every withdrawal waits your delay, 24 hours to 7 days.
- Instant moves go only to your own safe addresses.
- Settings changes wait the same delay.
- A freeze voids every pending request at once.
- Early release needs you and a guardian together.

Key	Lives in	Can
Owner	Seed Vault	Deposit, request, cancel
Guardian	Their phone	Cancel, freeze, check in
Sentinel	This phone's keystore	Only freeze and cancel

ARCHITECTURE

Two phones, one program. The chain holds the money and the rules.



ON-CHAIN PROGRAM

5 account types, 17 instructions, each with fixed signers.

Accounts (program-derived)

Vault	["vault", owner]: SOL savings, roles, delay, freeze, epoch
Vault token account	Savings in SKR or another SPL token; only the vault PDA signs
PendingWithdrawal	["withdrawal", vault, id]: amount, fixed destination, unlock time, epoch
PendingConfig	["config", vault]: proposed settings, apply time, epoch
StipendPool	["stipend", vault]: SKR mint, weekly rate, last claim per guardian

Instructions, by who must sign

Owner	init_vault, request_withdrawal, instant_withdraw (safe list only), propose_config, setup_stipend
Owner + guardian	expedite_withdrawal, lift_lockdown
Owner, guardian or sentinel	cancel_withdrawal, cancel_config, lockdown
Guardian	claim_stipend, guardian_heartbeat
Anyone	deposit_sol, deposit_token, fund_stipend; execute_withdrawal and apply_config only after the delay

Only three instructions move savings out: execute (after the delay, to the stored destination), expedite (owner and guardian together) and instant (safe list only). A freeze bumps the epoch, voiding every pending request.

SECURITY MODEL

The delay cannot be skipped, even with the owner's real key.

- The program sets the unlock time, not the caller.
- Payout goes only to the destination stored at request time.
- Instant payouts only reach safe addresses already on the list.
- Adding a safe address, shortening the delay or swapping a guardian waits the current delay, and any role can cancel it.
- A freeze voids every pending request; lifting it early needs owner and guardian together.

If this is compromised	They can	They cannot
The unlocked phone	Ask the wallet to sign a request; freeze	Skip the delay, or reach the owner key
The owner's key	Request, propose, pay out to existing safe addresses	Pay a new address before the delay; change settings faster
One guardian	Cancel and freeze	Move, redirect or release any savings
The sentinel key	Freeze and cancel	Move funds or change settings
Upgrade authority	Deploy new rules	Nothing, once made immutable (before mainnet)

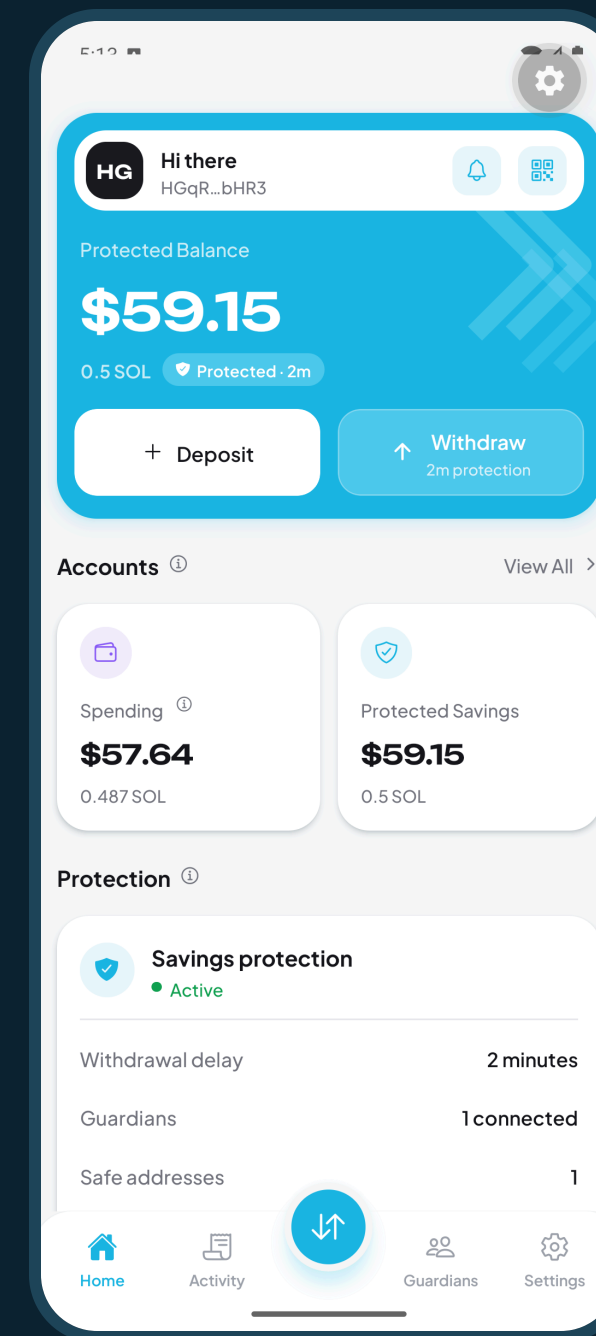
Known gap: a hostile guardian can keep re-freezing. Savings are held up, never taken; a per-role freeze cooldown is planned. Each rule has a test, and 22 of 22 pass on the deployed binary.

THE BACKUP PIN

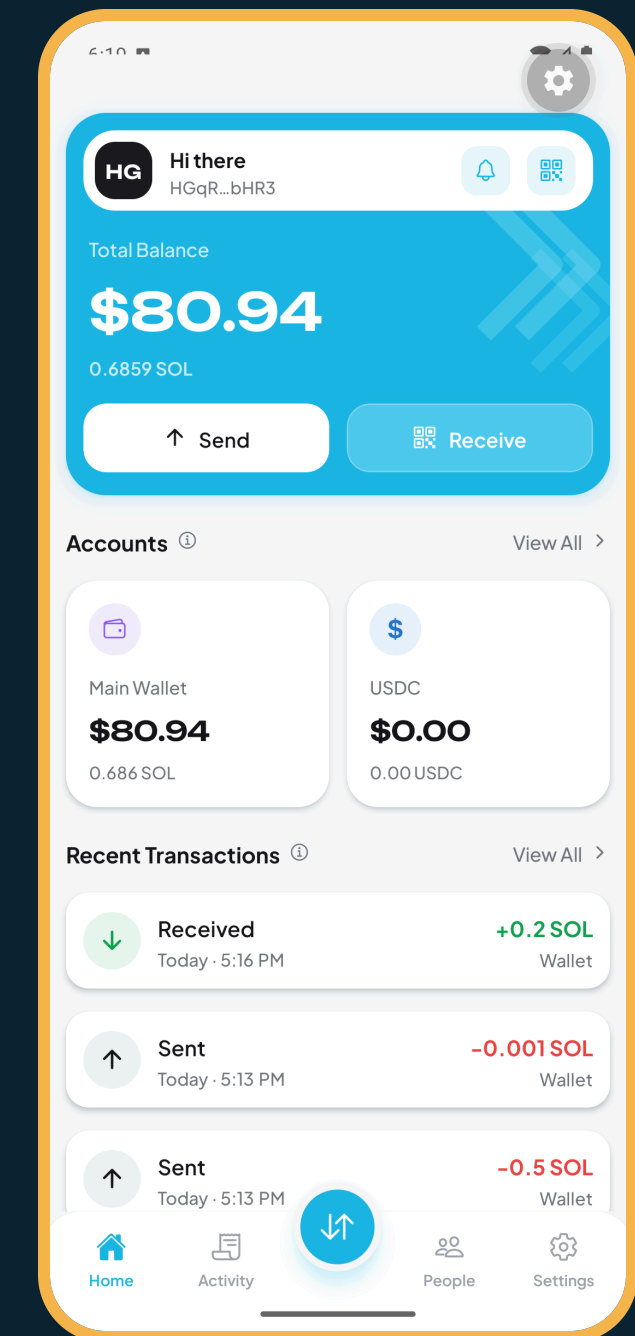
Two PINs. One opens an ordinary wallet.

Both look and behave the same. The backup PIN shows only your spending wallet, real balance and real history, while in the background:

- Savings freeze on-chain, signed by this phone's own key. No wallet prompt.
- Your emergency contact gets a text with your location.
- Any pending withdrawal is void for good.



Your PIN



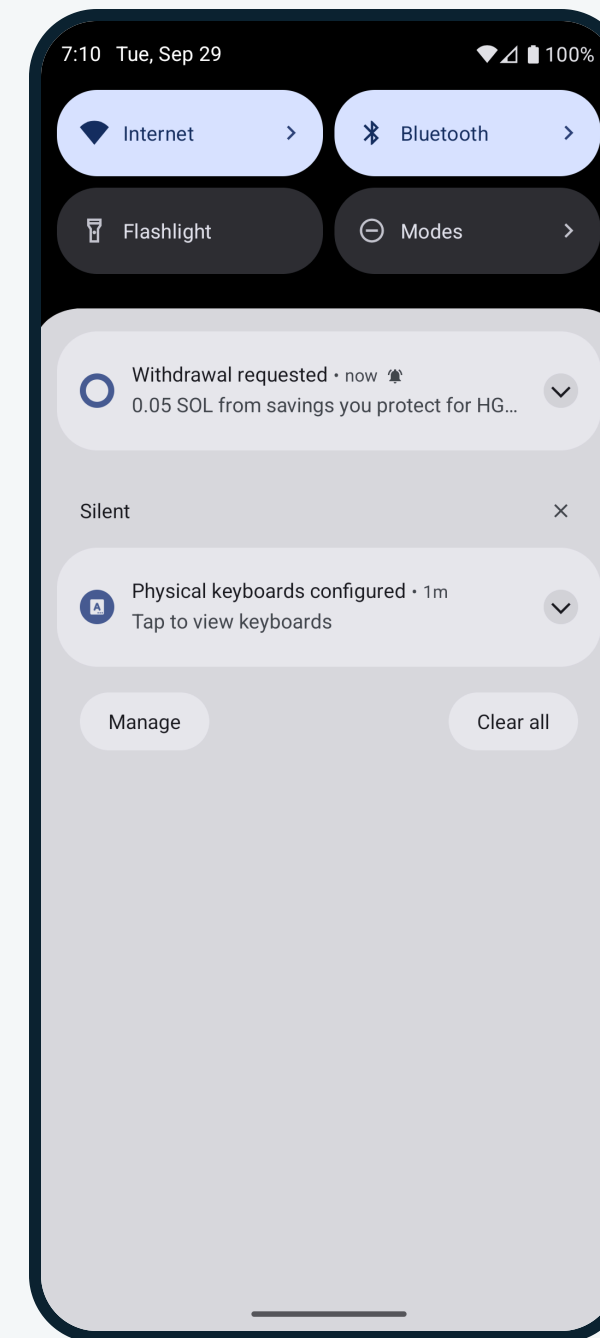
Backup PIN

GUARDIANS

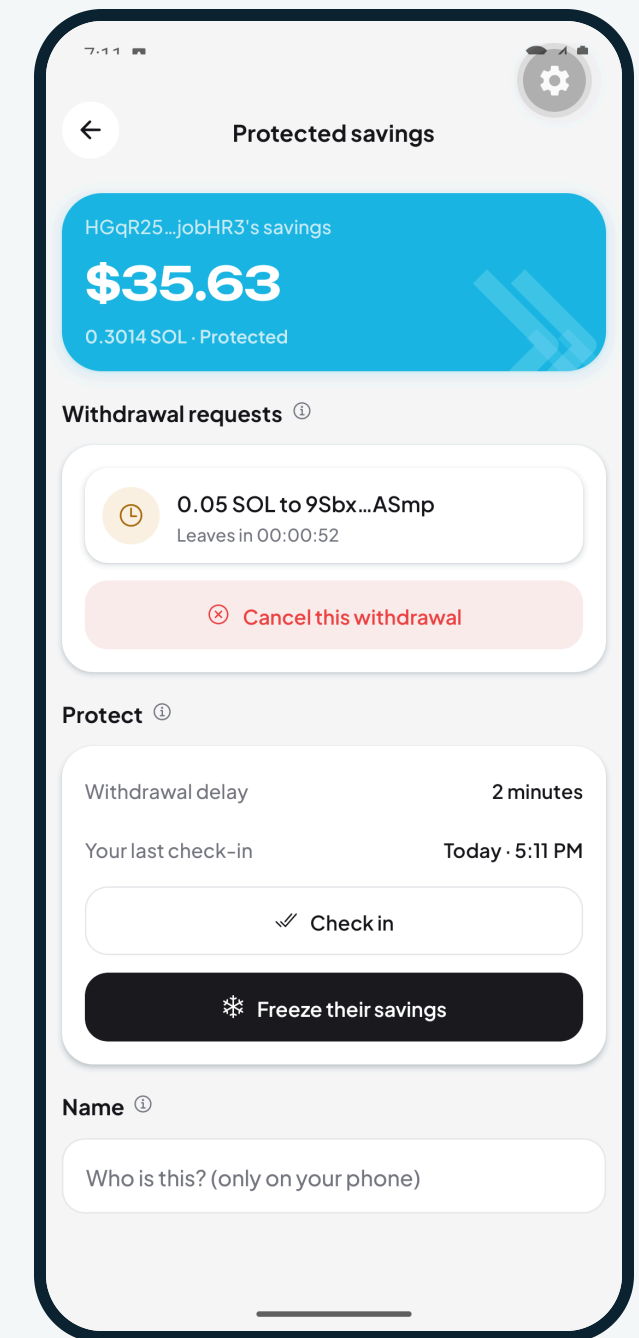
Someone you trust can stop it in time.

- Same app, their own wallet. Found automatically, no invite handshake.
- Alerted when a withdrawal is requested, or the backup PIN is used.
- Can cancel, freeze and check in. Can never take money out.

In testing, the alert arrived about 5 seconds after an outside request.



The alert



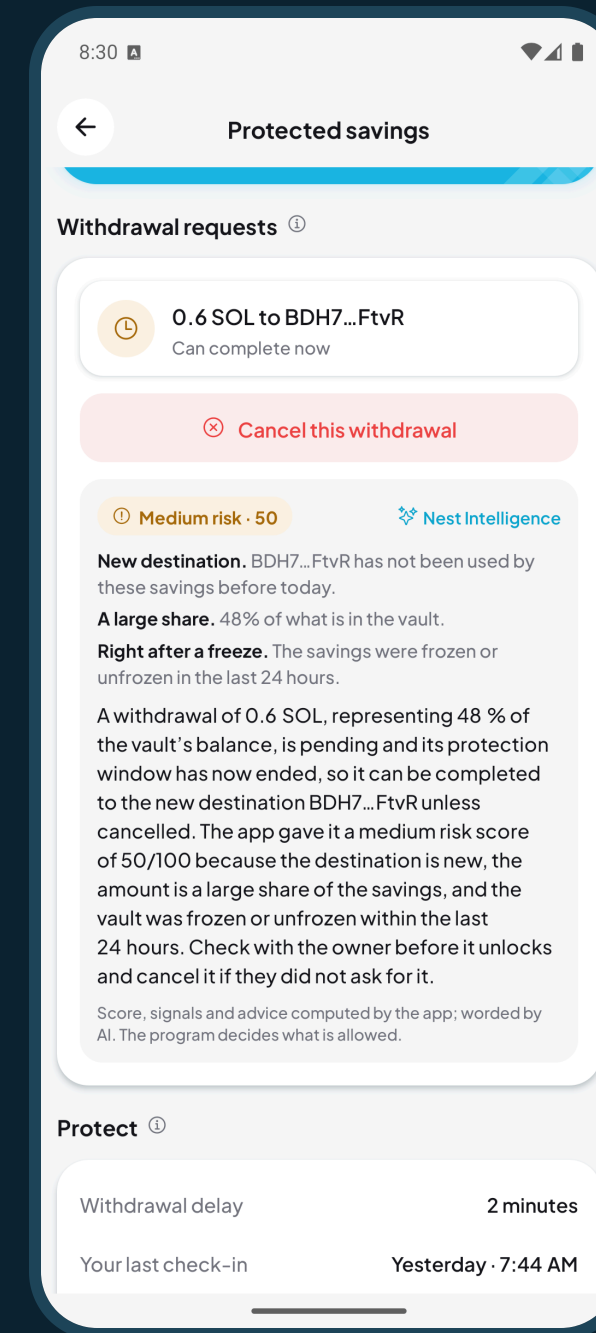
One tap to cancel

NEST INTELLIGENCE

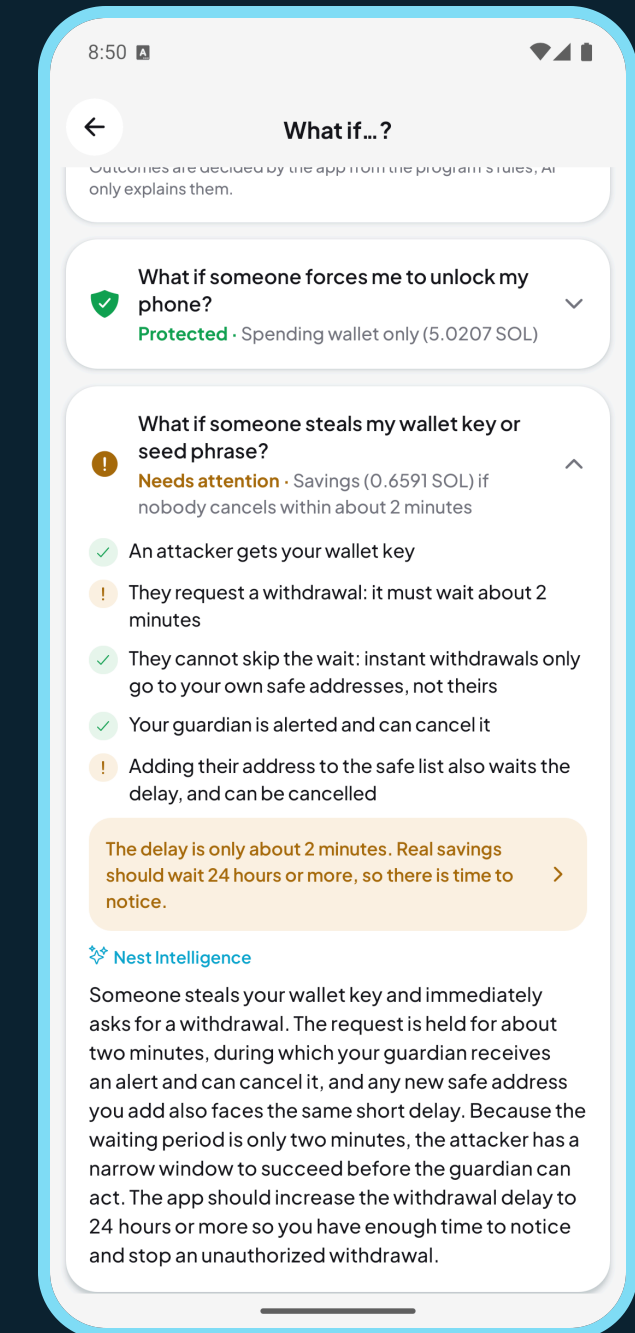
AI explains, scores and simulates. The program decides.

- **Risk scoring:** every withdrawal scored from on-chain signals; guardians get a briefing, and the alert says "High risk".
- **Transaction explainer:** what you are about to sign, in plain words, before the wallet opens.
- **What-if simulator:** seven attacks played against your real setup, with fixes and free-form questions.

Scores and outcomes are computed in code; Groq only words them.
No keys, addresses or location are sent.



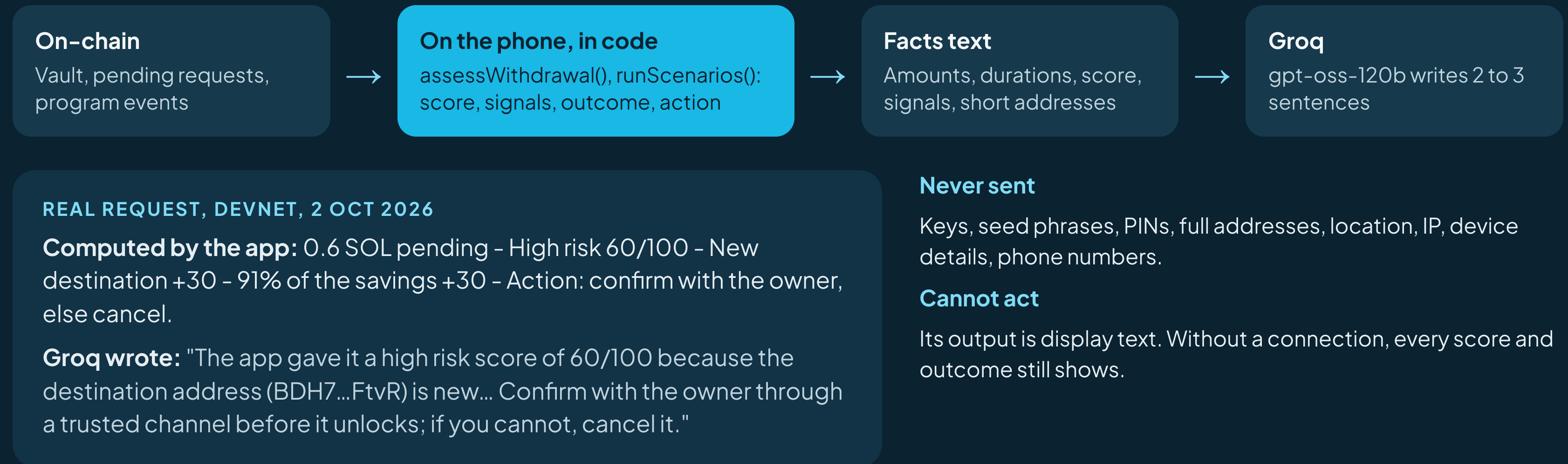
Guardian briefing



What-if simulator

AI DATA FLOW

The app decides. Groq only words what it decided.

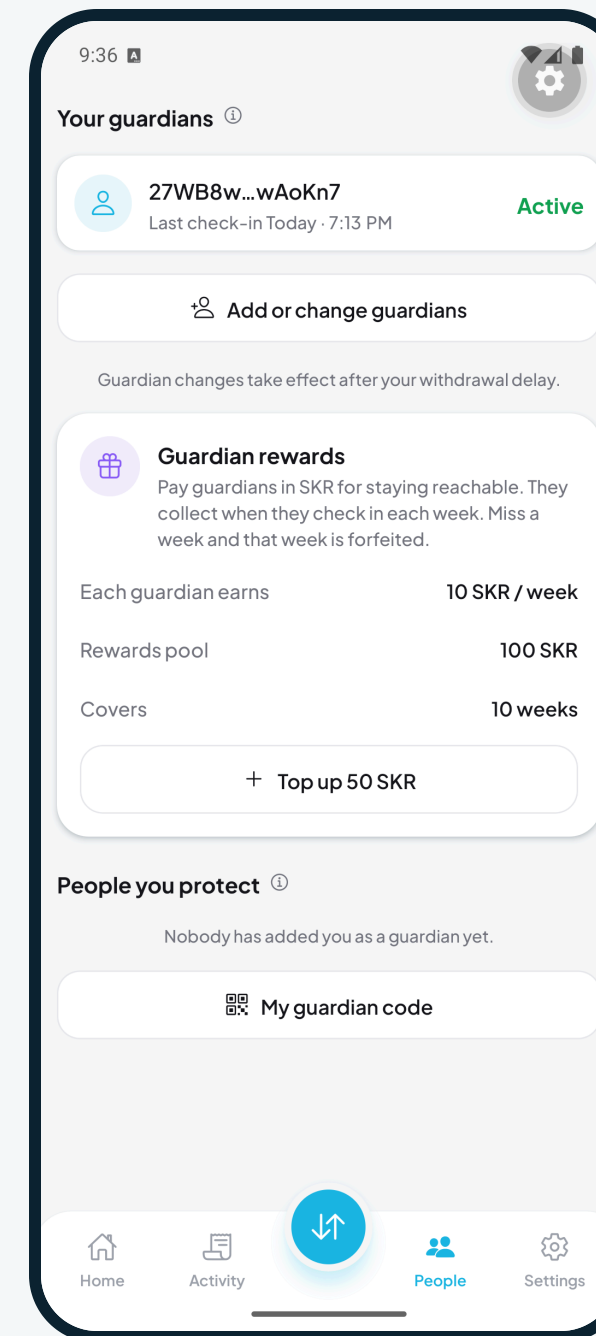


SKR

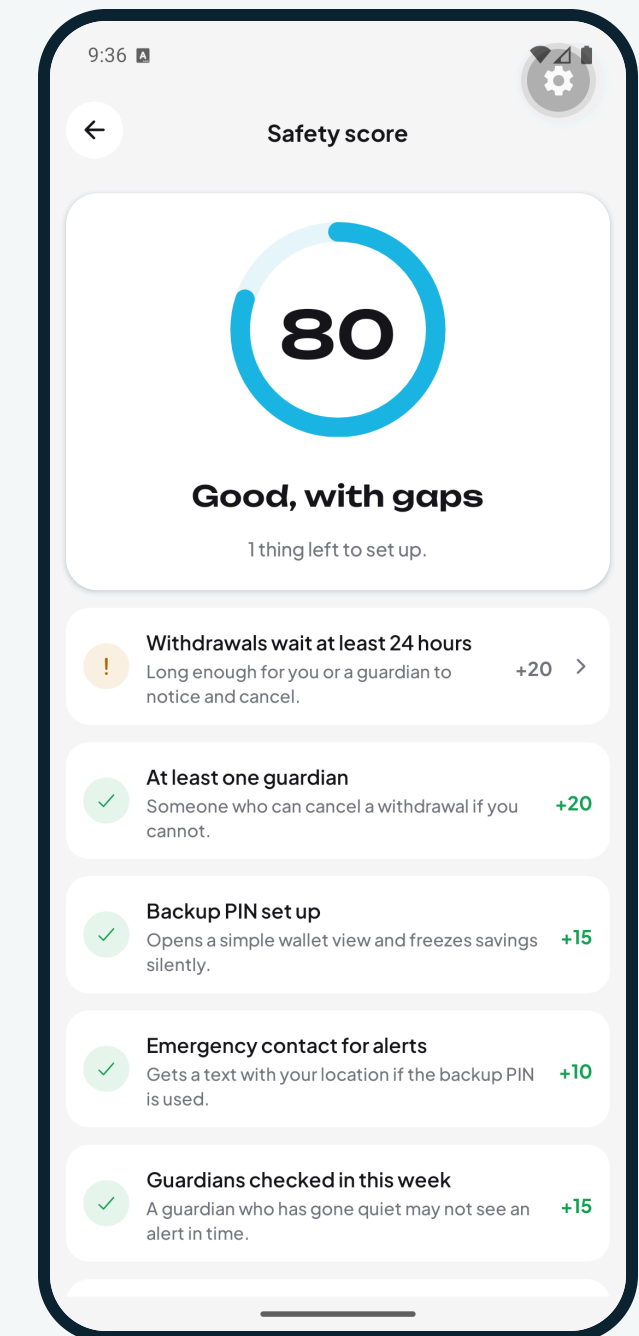
Guardians earn SKR for staying reachable.

- The owner funds a rewards pool in SKR.
- Guardians collect when they check in each week.
- Each claim covers at most 8 days, so a missed week is forfeited.
- The safety score shows gaps, from delay length to funded rewards.

Tested against SKR's real mainnet mint account. Mainnet is a config change, not a code change.



Rewards pool



Safety score

A full reward cycle, traced on devnet.

1

Pool funded

setup_stipend + fund_stipend:
5 tokens into the pool's own
account, at 1 token a week.

tx 5ZVcKv...BgHz1L

2

First check-in

claim_stipend pays 1 token (a
first claim counts as one week)
and records the heartbeat.

tx q2cKdV...WX4xoa

3

152 s later

Pays 251 base units: $1,000,000 \times 152 \div 604,800 = 251$, exactly
the formula.

tx 4q8Bf2...qWP5TJ

4

Heartbeat

guardian_last_seen
1790922408 → 1790922560,
read back from the vault.

same transaction

Rules the program enforces

$\text{paid} = \text{rate} \times \min(\text{time since last claim}, 8 \text{ days}) \div 1 \text{ week}$, never more
than the pool holds. Only current guardians can claim. The pool is
separate from the savings.

Mainnet switch

Devnet uses Circle's devnet USDC as the stand-in (same Token
program, 6 decimals). Mainnet sets the mint to SKR
(SKRbvo6Gf7GondiT3BbTfuRDPqLWei4j2Qy2NPGZhW3): a
config value, tested against SKR's real mint account.

BUILT FOR SEEKER

Built for the phone, not ported to it.

Seed Vault + MWA

The owner key never leaves Seed Vault. Every action is approved in the wallet sheet.

Sentinel key

A tighten-only key in Android Keystore signs the silent freeze.

Silent SMS

A small Kotlin module sends the alert with nothing on screen. Android only.

Real data only

The backup-PIN view shows the real wallet, so an explorer check matches.

Was this you?

An alert for any withdrawal this phone didn't request.

Native app

Expo and React Native with @solana/kit and a typed client generated from the IDL.

PROOF

Working on devnet today.

22 / 22

program tests pass on the deployed binary, including an attacker holding the owner's real key, and guardian rewards on SKR's real mainnet mint.

65 devnet transactions, decoded and linked in [docs/devnet-evidence.md](#):

- Vaults created, deposits, withdrawals after the delay
- Withdrawals cancelled by owner and guardian
- Backup-PIN freezes signed by the phone's own key
- Settings change waited out, then applied
- Rewards pool funded and collected on check-in

JUDGING CHECKLIST

Where to find each thing.

Criterion	Evidence	Where
Runs on Android, with MWA	APK on GitHub; every signature through Mobile Wallet Adapter; sentinel key in Android Keystore	GitHub release; use-vault.ts; sentinel.ts
Stickiness	Weekly guardian check-ins paid in SKR; risk alerts; safety score and What-if simulator	skr.md; simulator.ts
User experience	Backup PIN decoy, one-tap cancel, plain-language AI, faucet buttons and demo timers for judges	README: For judges
Innovation	Duress PIN + on-chain time lock + a key that can only tighten, so a coerced unlock gives nothing	threat-model.md
Demo	2:50 video of the flows on devnet	youtube.com/shorts/_GFJuXn5x9o
SKR integration	Rewards pool enforced on-chain; full cycle traced; tested on SKR's mainnet mint	skr-trace.md
Technical evidence	22 of 22 tests; deployed binary matches the repo; IDL on-chain; 65 decoded transactions	EVIDENCE.md
AI, honestly scoped	Scores computed in code; Groq words them; exact payload published	ai.md

What works today, and what comes next.

Implemented, on devnet

- Anchor vault: delays, safe list, freezes, delayed settings
- Android app with Mobile Wallet Adapter
- Backup PIN: silent freeze by the Keystore key, SMS, freeze report
- Guardians: discovery, risk-scored alerts, one-tap cancel
- SKR rewards, tested on SKR's real mainnet mint
- Nest Intelligence: risk scoring, briefings, What-if simulator
- 22 tests on the deployed binary, 65 devnet transactions

Planned

- Professional audit, then mainnet with real SKR
- Upgrade authority made immutable or a multisig
- Guardian alerts delivered in the background (push)
- AI through a server-side proxy, so no key ships in the app
- Owner can close a rewards pool and reclaim unspent SKR
- A freeze cooldown per role, so a hostile guardian cannot keep re-freezing
- Listing on the Solana dApp Store

TEXT VERSION – SPEAKER NOTES

1. Nest Finance

Nest Finance is a savings app for the Solana Seeker. It looks like an ordinary personal finance app, but your savings sit in an on-chain vault that nobody can rush, not even someone holding your unlocked phone.

2. A Seeker in your hand says you hold crypto.

Owning a Seeker tells people you hold crypto. If someone forces you to unlock it, or you sign one bad transaction, everything can leave in seconds, and nothing in a normal wallet slows that down.

3. Even with your real key, nobody can rush your savings.

Nest Finance makes that encounter pointless. Savings sit in a vault where every withdrawal waits. A second PIN opens an ordinary-looking wallet and quietly freezes the savings. And guardians, people you trust, can cancel a withdrawal before it lands.

4. The program enforces the rules, not the app.

An Anchor program on Solana holds the savings. Withdrawals wait a delay; the only instant path is to your own safe addresses; even settings changes wait. There are three keys: the owner in Seed Vault, guardians on their own phones, and a sentinel key in this phone's keystore that can only tighten, never move funds.

TEXT VERSION – SPEAKER NOTES

5. Two phones, one program. The chain holds the money and the rules.

The owner's phone runs the Nest Finance app. The owner's key stays in Seed Vault or their wallet and signs through Mobile Wallet Adapter. The app also holds a sentinel key in Android Keystore that can only freeze and cancel, so the backup PIN can freeze the vault with no wallet prompt and text an emergency contact. The savings and every rule live in the nest_vault program on Solana. Guardians run the same app on their own phones: it watches the vault, raises a risk-scored alert, lets them cancel or freeze with one tap, and pays a weekly SKR check-in. Groq sits off to the side: it receives facts text and returns wording, nothing else.

6. 5 account types, 17 instructions, each with fixed signers.

The program has five account types, all derived from the owner's address, and seventeen instructions, each with a fixed set of signers. The owner alone can request and propose; the owner and a guardian together can release early or lift a freeze; any role can cancel or freeze; guardians check in and claim; anyone can deposit, or run a withdrawal whose delay has passed. Only three instructions can move savings out, and a guardian or the sentinel cannot use any of them alone.

7. The delay cannot be skipped, even with the owner's real key.

This is why the delay holds even against someone with the owner's real key. The program sets the unlock time, the payout is pinned to the stored destination, instant payouts only reach addresses already on the safe list, and any change to the safe list, the delay or the guardians waits the current delay in plain view of the guardians. The table shows what each compromised key can and cannot do. One honest gap: a hostile guardian can keep re-freezing the vault. That holds savings up but never takes them, and a freeze cooldown is planned.

8. Two PINs. One opens an ordinary wallet.

If you're forced to open the app, enter your backup PIN. It opens what looks like a normal wallet with only your spending money, all real data. Behind the scenes the phone's own key freezes your savings with no wallet prompt, and your emergency contact gets a text with your location.

TEXT VERSION – SPEAKER NOTES

9. Someone you trust can stop it in time.

A guardian opens the same app with their own wallet, and Nest finds every vault that lists them. When a withdrawal is requested, maybe by a drainer, maybe under pressure, the guardian gets an alert and can cancel it before the delay runs out.

10. AI explains, scores and simulates. The program decides.

Nest Intelligence makes the security understandable. Every withdrawal is scored for risk from on-chain signals, like a destination never used before or most of the savings leaving at once, and guardians get a plain-language briefing, with the risk level right in the alert. Before signing, the owner sees what the transaction does. And the What-if simulator plays seven attacks against the owner's real setup and says what to fix. The principle: the app computes the scores and outcomes, AI only words them, and the Solana program decides what is allowed.

11. The app decides. Groq only words what it decided.

This is exactly what goes where. The app reads the vault and its pending requests from the chain, and plain code computes the risk score, the signals and the recommended action. Only a short facts text goes to Groq, which writes two or three sentences. Here is a real request from devnet: the app scored a 0.6 SOL withdrawal High, 60, because the destination was new and it was 91 percent of the savings; Groq only put that into words. Keys, PINs, full addresses and location are never sent, and the model can do nothing but show text.

12. Guardians earn SKR for staying reachable.

A guardian is only useful if they're reachable. Owners fund a pool in SKR, and guardians collect it when they check in each week. The program caps every claim at 8 days, so skipping a week forfeits that week. The safety score keeps owners honest about gaps. A program test loads SKR's actual mainnet mint account and runs the whole cycle, so moving to mainnet is a configuration change.

TEXT VERSION – SPEAKER NOTES

13. A full reward cycle, traced on devnet.

Here is one complete rewards cycle on devnet, read back from the chain after each step. The owner funds a pool. The guardian's first check-in pays one week. A second check-in 152 seconds later pays 251 base units, exactly what the formula gives, and both update the guardian's heartbeat on the vault. The 8-day cap is checked by a program test that skips ahead 30 days. On mainnet the same program runs with the SKR mint.

14. Built for the phone, not ported to it.

This only works on a phone. The owner key stays in Seed Vault, a tighten-only sentinel key lives in the Android keystore, and a small native module sends the alert text silently, something iOS apps can't do.

15. Working on devnet today.

Everything you've seen runs on devnet today. The program has 22 tests, run against the exact binary deployed on devnet: one has an attacker holding the owner's real key try every route to the money, and one runs guardian rewards on SKR's real mainnet mint account. On devnet there are 65 decoded transactions on the demo vaults, each linked in the repository's evidence page.

16. Where to find each thing.

For judges, this maps each criterion to its evidence: the APK and the Mobile Wallet Adapter code, the weekly SKR check-ins that bring guardians back, the backup-PIN experience, the on-chain design behind it, the demo video, the SKR trace, the tests and deployment proof, and the AI payload. Everything is in the repository, starting from EVIDENCE.md.

TEXT VERSION – SPEAKER NOTES

17. What works today, and what comes next.

Here is exactly what is built and what is not. Built and running on devnet today: the vault with all its rules, the Android app, the backup PIN with its silent freeze and alert, guardians with risk-scored alerts, SKR rewards tested against the real SKR mint, and Nest Intelligence. Next: an audit, mainnet with real SKR, a locked upgrade authority, background push alerts for guardians, moving the AI key behind a server, and a dApp Store listing. Nest Finance: savings that can't be rushed.

Nest Finance

Nest Finance lock screen on a phone

SCREEN TEXT (OCR, MACHINE-READ)

9:33
Nest Finance
Enter your PIN

Unlock to see your accounts
0000

Two PINs. One opens an ordinary wallet.

Normal view with protected savings

SCREEN TEXT (OCR, MACHINE-READ)

r. -10
Hi there
HGqR...bHR3
Protected Balance
\$59.15
0.5 SOL Protected • 2m
+ Deposit
Accounts
6
Spending O
\$57.64
0.487 SOL
Protection
Savings protection
Active
Withdrawal delay

Guardians
Safe addresses
Withdraw
2m protection
View All >
Protected Savings
\$59.15
0.5 SOL
2 minutes
1 connected
oo
Home
Activity
Guardians
Settings

Backup PIN view showing only the spending wallet

SCREEN TEXT (OCR, MACHINE-READ)

Hi there
HGqR...bHR3
Total Balance
\$80.94
0.6859 SOL
'Tx Send
Accounts
6
Main Wallet
\$80.94
0.686 SOL
Recent Transactions
Received
Today • 5:16 PM
Sent
Today • 5:13 PM
Sent
Today • 5:13 PM

Receive
View All >
USDC
\$0.00
0.00USDC
View All >
+0.2SOL
Wallet
-0.001 SOL
Home
Activity
oo
People
Wallet
-0.5SOL
Wallet
Settings

Someone you trust can stop it in time.

Guardian alert: withdrawal requested

SCREEN TEXT (OCR, MACHINE-READ)

7:10 Tue, sep 29

Internet

Flashlight

Bluetooth

Modes

Withdrawal requested • now n

0.05 SOL from savings you protect for

HG...

Silent

Physical keyboards configured • 1m

Tap to view keyboards

Manage

Clear all

Guardian view with a cancel button

SCREEN TEXT (OCR, MACHINE-READ)

04 •

Protected savings

HGqR25...jobHR3's savings

\$35,63

0.3014 SOL • Protected

Withdrawal requests

0.05 SOL to 9Sbx...ASmp

O

Leaves in

@ Cancel this withdrawal

Protect

Withdrawal delay

Your last check-in

V/ Checkin

2 minutes

Today. 5:11 PM

* Freeze their savings

Name

Who is this? (only on your phone)

AI explains, scores and simulates. The program decides.

Guardian view of a pending 0.6 SOL withdrawal: Medium risk 50, signals New destination, A large share, Right after a freeze, and an AI briefing advising to check with the owner

SCREEN TEXT (OCR, MACHINE-READ)

8:30	to the new destination BDH7... FtvR
Protected savings	unless
Withdrawal requests	cancelled. The app gave it a medium risk
0.6 SOL to BDH7...FtvR	score
Can complete now	of 50/100 because the destination is
@ Cancel this withdrawal	new, the
O Medium risk 50	amount is a large share of the savings,
Nest Intelligence	and the
New destination. BDH7... FtvR has not	vault was frozen or unfrozen within the
been used by	last
these savings before today.	24 hours. Check with the owner before it
A large share. 48% of what is in the vault.	unlocks
Right after a freeze. The savings were	and cancel it if they did not ask for it.
frozen or	Score, signals and advice computed by
unfrozen in the last 24 hours.	the app; worded by
A withdrawal of 0.6 SOL, representing 48	AI. The program decides what is allowed.
% of	Protect
the vault's balance, is pending and its	Withdrawal delay
protection	Your last check-in
window has now ended, so it can be	2 minutes
completed	Yesterday 7:44 AM

What-if simulator: someone steals my wallet key, needs attention, with each step, the fix to lengthen the delay, and an AI walkthrough

SCREEN TEXT (OCR, MACHINE-READ)

8:50	savings
What if... ?	should wait 24 hours or more, so there is
a' C uy app HI	time to
only explains them.	notice.
What if someone forces me to unlock my	Nest Intelligence
phone?	Someone steals your wallet key and
Protected • Spending wallet only	immediately
(5.0207 SOL)	asks for a withdrawal. The request is held
What if someone steals my wallet key or	for about
seed phrase?	two minutes, during which your guardian
O	receives
Needs attention • Savings (0.6591 SOL) if	an alert and can cancel it, and any new
nobody cancels within about 2 minutes	safe address
An attacker gets your wallet key	you add also faces the same short delay.
They request a withdrawal: it must wait	Because the
about 2	waiting period is only two minutes, the
minutes	attacker has a
They cannot skip the wait: instant	narrow window to succeed before the
withdrawals only	guardian can
go to your own safe addresses, not theirs	act. The app should increase the
Your guardian is alerted and can cancel it	withdrawal delay to
Adding their address to the safe list also	24 hours or more so you have enough
waits the	time to notice
delay, and can be cancelled	and stop an unauthorized withdrawal.
The delay is only about 2 minutes. Real	

Guardians earn SKR for staying reachable.

Guardian rewards card: 10 SKR per week

SCREEN TEXT (OCR, MACHINE-READ)

9:36	Each guardian earns
Your guardians	Rewards pool
o	Covers
27WB8w...wAoKn7	+ Topup50SKR
Last check-in Today 7:13 PM	People you protect
+0 Add or change guardians	10 SKR/week
Active	100 SKR
Guardian changes take effect after your withdrawal delay.	10 weeks
Guardian rewards	Nobody has added you as a guardian yet.
Pay guardians in SKR for staying reachable. They	My guardian code
collect when they check in each week.	Home
Miss a	Activity
week and that week is forfeited.	People
	Settings

Safety score of 80

SCREEN TEXT (OCR, MACHINE-READ)

9:36	savings
Safety score	silently.
80	Emergency contact for alerts
Good, with gaps	Gets a text with your location if the
1 thing left to set up.	backup PIN
Withdrawals wait at least 24 hours	is used.
Long enough for you or a guardian to notice and cancel.	Guardians checked in this week
At least one guardian	A guardian who has gone quiet may not
+20	see an
Someone who can cancel a withdrawal if you	alert in time.
cannot.	+20
Backup PIN set up	+15
Opens a simple wallet view and freezes	+10
	+15